

User stories completed this sprint:

SCRUM-55: As a user, I want to opt in/out of email notifications in my settings page, so I only receive updates that matter to me.

SCRUM-46: As a user, I want to be able to upload photos corresponding to specific beaches/bodies of water, so I can further incentivize the community aspect of the app

SCRUM-58: As a user, I want to be able to enable or disable emails, so I don't have to receive emails I don't want

SCRUM-27: As a user, I want to be able to get contribution points, so that I can be rewarded for entering data

SCRUM-42: As a user, I want to be able to access a 'community' tab so I can see insights about popular water destinations in my area

SCRUM-45: As a user, I want to be able to identify key points in the map that correlate with 'Points of Interest' so I can identify meaningful points on the heatmap tool

SCRUM-28: As a user, I want to be able to have my current location tracked, so that I can easily enter data

SCRUM-51: As a user, I want to receive email updates, so that I can be up to date on new features

SCRUM-50: As a user, I want to be able to quickly upload temperature by the system knowing my location, so I don't have to waste time manually adding latitude/longitude values

SCRUM-32: As a user, I want to be able to view a history of average temperatures for a specific water body when I click on the map.

SCRUM-57: As an admin, I want to sort user input data by the date, so I can easily verify data for correctness

SCRUM-61: As a user, I want to be able to view other users' photo submissions, and interact with their photos by liking/unliking them

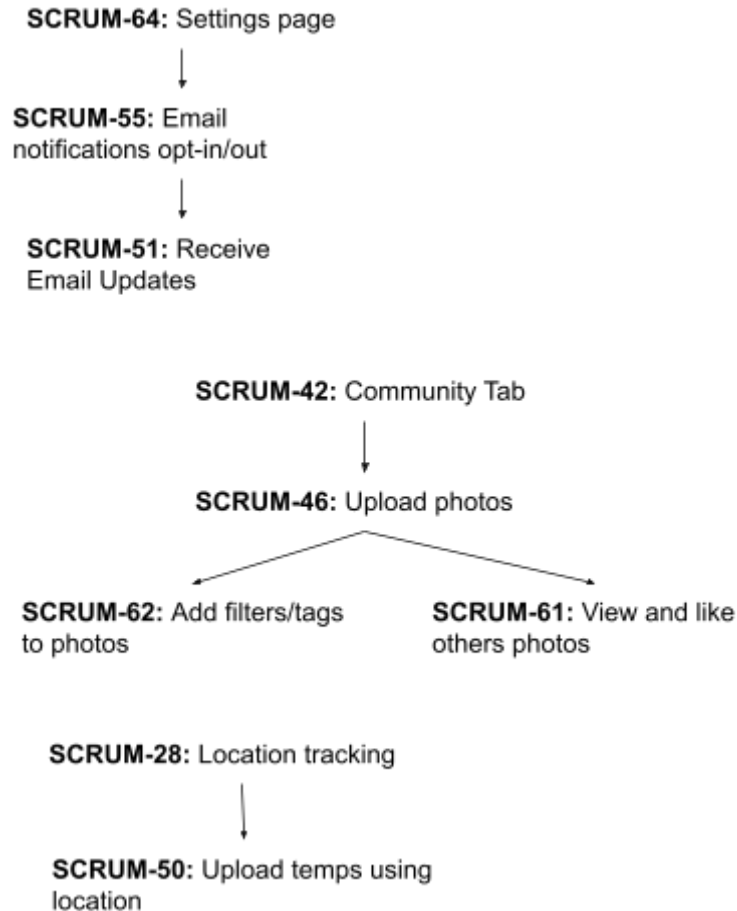
SCRUM-62: As a user, I want to be able to add filters/tags to my photo uploads and sort them by these tags when viewing them

SCRUM-64: As a user, I want to be able to access a settings page allows me to update my display name, delete account, and manage my personal info

SCRUM-63 As an admin, I want to be able to view user inputs in a filterable, and

manageable way to easily decide whether to reject them

Network Diagram



Dependencies

Settings and Notifications Flow

- **SCRUM-64:** Settings page lays the foundation for user-specific preferences.
- This enables **SCRUM-55:** Email notifications opt-in/out, which in turn enables **SCRUM-51:** Receive Email Updates.
- The flow builds from general user preferences to specific notification behavior.

Community and Photo Sharing Flow

- **SCRUM-42:** Community Tab must be implemented first to provide access to community features.

- It enables **SCRUM-46**: Upload photos, which then branches into two enhancements:
 - **SCRUM-62**: Add filters/tags to photos
 - **SCRUM-61**: View and like others' photos
- Each of these features relies on photo upload functionality being complete.

Location-Based Upload Flow

- **SCRUM-28**: Location tracking is a prerequisite for any geotagged content.
- It enables **SCRUM-50**: Upload temperatures using location, which incorporates environmental data into uploads.

Critical Path

SCRUM-42 -> SCRUM-46 -> [SCRUM-62 and SCRUM-61]

This flow begins with creating the community hub (SCRUM-42), followed by enabling photo uploads (SCRUM-46), and then expanding the experience through filters/tags (SCRUM-62) and viewing/liking photos (SCRUM-61). Any delay in photo upload functionality would have blocked both aesthetic customization and social engagement, which are key to the community experience.

Other paths (settings/email and location tracking) are functionally important but not on the critical path, as they do not directly gate key engagement features like sharing and interaction.

Risk Areas (things that could've gone wrong)

A few key risks were present across the three feature flows:

- **Community Flow Risks**
 - Failure in SCRUM-46 (photo upload) would block both visual tagging and user interaction entirely, making this a high-risk node.
 - Inconsistent image handling or performance issues could also degrade the perceived reliability of the platform.
- **Settings/Email Flow Risks**
 - Email notification logic (SCRUM-55) depended on correctly persisting opt-in preferences from the settings page (SCRUM-64). Incorrect or delayed delivery in SCRUM-51 could lead to poor communication with users.
- **Location Upload Risks**
 - Location tracking (SCRUM-28) had platform-level dependencies (permissions, device support).

- Errors in geolocation accuracy would have led to unreliable or misleading temperature uploads in SCRUM-50.

To mitigate these risks, the team prioritized stable backend integration early in the sprint, introduced robust error handling (especially around media and location), and staggered implementation so high-risk features were tested before branching enhancements.